Anion Receptor Addition as a Mechanism for Coordination Entropy Reduction: LiFSI in DOL/Trimethyl Borate as a Designed Approach to SCE* = 1.466 from Above

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Abstract

The SCE Framework establishes $\mathrm{SCE}^* = 1.466$ as the Shannon Coordination Entropy of the Li^+ first solvation shell that simultaneously maximises room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) in lithium metal batteries. Navigation to this fixed point requires two distinct levers depending on the position of the base electrolyte relative to SCE^*: systems below SCE^* require entropy addition; systems above SCE^* require entropy reduction.

This paper addresses the second case. We identify 1,3-dioxolane (DOL) as a base solvent whose Li^+ solvation shell sits above SCE^* ($\mathrm{SCE}_{\mathrm{DOL}} \approx 1.60$–$1.65$): the shell is too disordered, multiple coordination configurations are comparably populated, and $\mathrm{CE}_{\mathrm{RT}}$ is suppressed by SEI inconsistency arising from excessive configuration diversity.

We introduce the **anion receptor mechanism** as the correct intervention: trimethyl borate (TMB) coordinates the FSI^- anion preferentially over Li^+, withdrawing it from competitive participation in the Li^+ primary shell. This selectively removes the anion-mediated coordination configurations (CIP and AGG species) from the shell population without altering the solvent geometry or introducing a new coordination class. The population contracts: $p_{\max}$ rises, n_{sig} falls, and SCE decreases toward SCE^*.

The mechanism is geometrically asymmetric with respect to ESP-matched co-solvent addition: where ESP matching raises SCE by splitting the population across a new structural class, anion reception lowers SCE by collapsing one existing structural contributor without replacing it. These are distinct, orthogonal levers acting on opposite sides of SCE^*.

Candidate 3 (LiFSI 1.0 M in DOL/TMB 4:1 v/v, target $\mathrm{SCE} = 1.466 \pm 0.05$) is derived with pre-registered falsification conditions at three levels of specificity. A molecular search across the boron ester class confirms TMB as the canonical anion receptor within accessible commercial chemistry. The paper establishes the anion receptor mechanism as a general, computable, and experimentally actionable principle for approaching SCE^* from above in any electrolyte system whose base solvation shell is too disordered.

Keywords: lithium metal battery, electrolyte design, anion receptor, trimethyl borate, DOL, coordination entropy, solvation structure, SCE*, fixed point, CIP, AGG, SEI

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1 Introduction

1.1 The Fixed Point and the Two Navigation Directions

The SCE Framework establishes the following result: there exists a unique value of the Shannon Coordination Entropy of the Li^+ first solvation shell,

$$\mathrm{SCE}^* = 1.466, \tag{1}$$

at which combined RT and LT Coulombic efficiency is simultaneously maximised. This result is derived analytically from the intersection of two opposing empirical gradients (RT performance decreases with SCE; LT performance increases with SCE) and confirmed across a 21-system published dataset ($R^2 = 0.828$, $p = 4.5\times10^{-6}$). The full derivation, empirical confirmation, and pre-registration record are available at https://github.com/Eric-Robert-Lawson/attractor-oncology.

Navigation to SCE* is directional. The position of the base electrolyte relative to the fixed point determines the correct intervention:

- **If** $\mathrm{SCE} < \mathrm{SCE}^*$**:** the shell is too ordered. The dominant configuration fraction $p_{\max}$ is too high. The correct intervention adds structural diversity to the shell, raising SCE toward the fixed point.
- **If** $\mathrm{SCE} > \mathrm{SCE}^*$**:** the shell is too disordered. Multiple configurations are comparably populated. The correct intervention removes a source of coordination diversity, contracting the population and lowering SCE toward the fixed point.

The two cases require fundamentally different chemical strategies. This paper addresses the second case exclusively.

1.2 Why DOL Sits Above SCE*

1,3-Dioxolane (DOL) is a cyclic ether widely used in lithium metal battery electrolytes. Its ring structure provides flexibility in the coordination geometry: the two ether oxygens can adopt multiple configurations relative to Li^+, and the ring conformational freedom creates a broader distribution of accessible coordination states than linear ethers.

In LiFSI/DOL systems at standard salt concentrations (∼1 M), the Li^+ first-shell population is distributed across multiple configurations: SSIP (solvent-separated ion pair), CIP (contact ion pair, one FSI^- in the shell), and mixed solvent/anion configurations. No single configuration dominates strongly. This is the signature of a shell above SCE*: diverse, flexible, multiple comparably populated configurations.

The empirical consequence is predictable from the SCE Framework: good LT performance (multiple low-barrier desolvation pathways available) but suppressed RT performance (SEI inconsistency from diverse decomposition chemistry). The system requires not more diversity but less.

1.3 What Has Not Been Done Before

Prior strategies for modifying DOL-based electrolytes include:

- Salt concentration increase (drives system into AGG-dominant Regime 2, different mechanism).
- Co-solvent addition (typically DME, which adds an additional coordination class and may raise SCE further rather than lower it).

- Additive packages targeting SEI chemistry directly rather than solvation geometry.
- Fluorinated ether additions to reduce solvent donor strength.

None of these strategies identify the source of the RT performance problem as excessive coordination entropy and apply an anion receptor to selectively remove the anion-mediated coordination configurations that contribute the entropy excess. The anion receptor mechanism introduced here is absent from the prior design literature as a rationally derived intervention targeting SCE reduction.

1.4 Structure of This Paper

Section 2 develops the theoretical basis of the anion receptor mechanism. Section 3 characterises the DOL baseline and its position above SCE*. Section 4 identifies TMB as the canonical anion receptor and develops the population contraction prediction. Section 5 states the candidate formulation and pre-registered falsification conditions. Section 6 reports the molecular search confirming TMB as the canonical boron ester anion receptor. Section 7 develops the geometric asymmetry between anion reception and ESP matching as orthogonal levers. Section 8 generalises the principle. Section **??** discusses implications and limitations.

2 The Anion Receptor Mechanism

2.1 The Source of Excess SCE in DOL-Based Systems

In a DOL/LiFSI electrolyte at 1 M salt concentration, the Li^+ first-shell population contains contributions from three coordination classes:

1. **SSIP configurations:** Li^+ coordinated exclusively by DOL molecules. FSI^- is excluded from the primary shell. Desolvation geometry: solvent-only, consistent, relatively low-barrier.
2. **CIP configurations:** Li^+ coordinated by a mixture of DOL and one FSI^- anion. The anion participates directly in the first shell. Desolvation geometry: mixed solvent/anion, distinct from SSIP geometry, higher barrier.
3. **Mixed/minority configurations:** Higher-order species (AGG, multi-anion contact) at low but non-negligible populations.

The CIP and mixed configurations are the source of the entropy excess. In a DOL system, the FSI^- anion is not strongly excluded from the Li^+ primary shell. At 1 M, FSI^- activity is sufficient to occupy a non-negligible CIP fraction. This CIP fraction is a distinct coordination class from the SSIP configurations and contributes directly to the Shannon entropy of the population distribution.

Definition 2.1 (Anion-mediated entropy contribution)**.** In a salt/solvent system where the anion participates in the Li^+ primary solvation shell with fractional population p_{CIP}, the anion-mediated entropy contribution is:

$$\Delta\mathrm{SCE}_{\mathrm{anion}} = -p_{\mathrm{CIP}} \ln p_{\mathrm{CIP}} - \sum_{i\in\mathrm{SSIP}} p_i \ln p_i \Big|_{\text{with CIP}} + \sum_{i\in\mathrm{SSIP}} p_i \ln p_i \Big|_{\text{without CIP}} \tag{2}$$

where the difference captures both the direct entropy of the CIP population and the indirect effect of CIP presence on the remaining SSIP fractions (which must sum to $1 - p_{\mathrm{CIP}}$ rather than 1).

Remark 2.1. $\Delta\mathrm{SCE}_{\mathrm{anion}} > 0$ whenever $p_{\mathrm{CIP}} > 0$. Removing the CIP class entirely (by intercepting FSI^- before it enters the Li^+ shell) reduces SCE by $\Delta\mathrm{SCE}_{\mathrm{anion}}$ and simultaneously redistributes the remaining SSIP population to sum to unity, further narrowing the distribution. The net effect is a contraction of the population toward fewer, more dominant configurations. SCE falls.

2.2 How an Anion Receptor Achieves This

An anion receptor is a Lewis acid molecule that coordinates the FSI^- anion with sufficient affinity to compete with Li^+ for FSI^- binding. When present in solution, the anion receptor:

1. Intercepts FSI^- before it enters the Li^+ primary shell.
2. Reduces the free FSI^- activity in solution.
3. Shifts the SSIP/CIP equilibrium toward SSIP.
4. Reduces p_{CIP} without altering the solvent coordination geometry.
5. Contracts the population distribution toward SSIP-dominated configurations.
6. Lowers SCE toward SCE*.

The critical feature of the mechanism is what it does *not* do: it does not introduce a new coordination class. The anion receptor molecule does not itself coordinate Li^+ to any significant degree. It operates exclusively on the anion, not on the Li^+ shell geometry. The shell contracts without acquiring new structural complexity.

Proposition 2.1 (Anion receptor design rule)**.** *An effective anion receptor for SCE reduction must satisfy:*

(i) Sufficient Lewis acidity to coordinate FSI^- in competition with Li^+ (binding free energy for FSI^- comparable to or exceeding the Li^+–FSI^- CIP binding energy).

(ii) Insufficient Lewis acidity to coordinate Li^+ directly (the receptor must not enter the Li^+ primary shell and introduce a new coordination class, which would raise SCE rather than lower it).

(iii) Chemical stability at Li metal potentials.

(iv) Liquid or soluble at operating temperature.

When all four conditions are satisfied, the receptor selectively intercepts FSI^-, collapses the CIP population, and lowers SCE without adding structural complexity.

3 The DOL Baseline and Its Position Above SCE*

3.1 SCE Characterisation of the DOL Baseline

From the confirmed dataset and derivation record archived at https://github.com/Eric-Robert-Lawson/attractor-oncology:

Table 1: DOL-based electrolyte baseline characterisation. SCE values computed from published coordination geometry populations. CE values from published experimental measurements. The DOL baseline sits above $SCE^* = 1.466$ in the dataset, with a required correction of $\Delta SCE \approx -0.15$ to reach the fixed point.

System	SCE	CE_{RT} (%)	CE_{LT} (%)	LT (°C)	Regime
LiFSI 1M DOL (pure)	∼1.60–1.65	∼81	∼68	−20	R1 est.
LiFSI 1M DOL/DME 1:1	1.27	98.9	27.5	−20	R1
LiFSI 1M DOL (target ratio)	1.466*	pred. 89	pred. 60	−20	R1

* Target achieved by TMB addition; pred. = predicted from confirmed framework equations.

Remark 3.1. The DOL/DME 1:1 system (S01 in the master dataset of Preprint 1) is informative precisely because it is a *negative* example of the correct intervention. DME addition to DOL does not lower SCE; it lowers SCE dramatically (from ∼1.62 to 1.27) by introducing a dominant DME coordination class that overwhelms the DOL diversity. The system overshoots SCE^* in the downward direction, producing excellent RT performance but collapsed LT performance ($CE_{LT} = 27.5\%$). This is the signature of a shell that has been overcorrected past the fixed point from above.

The DOL/DME 1:1 result is the strongest empirical evidence that DOL sits above SCE^* and that the correct intervention is a *gentle* contraction of the population — not the aggressive displacement that DME addition produces. TMB achieves the gentle contraction by operating on the anion rather than the solvent.

3.2 The Failure Mode Produced by Overcorrection

The framework predicts two distinct failure modes for systems near SCE^*:

Failure Mode A (undercorrection, SCE still above SCE^*): CE_{RT} remains suppressed. CIP configurations persist in the shell. SEI is inconsistent. The anion receptor dose was insufficient.

Failure Mode B (overcorrection, SCE driven below SCE^*): CE_{RT} recovers but CE_{LT} collapses. The shell is now too ordered. Only one dominant configuration exists. Desolvation at low temperature is blocked by the single rigid pathway. DOL/DME 1:1 is the canonical empirical example of this failure mode.

The falsification structure for Candidate 3 is designed to detect both failure modes: if CE_{RT} does not improve, Mode A is suspected. If CE_{LT} collapses while CE_{RT} improves, Mode B has occurred and the TMB dose is too high.

4 TMB as the Canonical Anion Receptor

4.1 Why TMB Satisfies the Anion Receptor Design Rule

Trimethyl borate (TMB, $B(OCH_3)_3$) is a boron ester with the following properties relevant to the anion receptor design rule:

TMB satisfies all four conditions of the anion receptor design rule:

(i) **Sufficient Lewis acidity for FSI^-:** The boron centre in TMB is electrophilic and interacts with the nitrogen or oxygen of FSI^-. This interaction is confirmed in the literature and is the basis for TMB's use as a flame retardant additive in lithium battery electrolytes.

Table 2: TMB properties relevant to the anion receptor mechanism. All values from primary sources cited.

Property	Value	Source
Melting point	$-29.3\,^\circ$C	CRC Handbook [6]
Boiling point	$68.7\,^\circ$C	CRC Handbook [6]
Lewis acidity (FIA, gas phase)	$\sim$340 kJ mol^{-1}	[4]
Li^+ coordination	Not observed (too weak)	[3]
FSI^- coordination	Confirmed (boron–N or boron–O interaction)	[9]
Electrochemical stability	Stable > 0 V vs. Li/Li^+	[5]
Availability	Commercial (Sigma-Aldrich, TCI)	—

(ii) **Insufficient Lewis acidity for Li^+:** TMB does not coordinate Li^+ directly. The boron centre is Lewis acidic, not basic; it has no lone pairs to donate to Li^+. TMB therefore does not enter the Li^+ primary shell and does not introduce a new coordination class.

(iii) **Chemical stability at Li metal potentials:** Electrochemical stability confirmed above 0 V vs. Li/Li^+.

(iv) **Liquid at operating temperature:** $T_m = -29.3\,^\circ$C, liquid at $-20\,^\circ$C (primary LT target).

4.2 The Population Contraction Mechanism in Detail

In LiFSI/DOL at 1 M without TMB, the FSI^- anion participates in the Li^+ primary shell at a CIP fraction estimated at $p_{\text{CIP}} \approx 0.15$–$0.25$ (consistent with Raman and MD data for similar systems [8, 1]). The population is distributed across SSIP, CIP, and minor mixed configurations. SCE ≈ 1.60–1.65.

As TMB is added:

1. TMB coordinates FSI^- in solution, forming a TMB–FSI^- complex.
2. Free FSI^- activity falls.
3. The SSIP/CIP equilibrium shifts toward SSIP: p_{CIP} falls.
4. The Li^+ shell population redistributes: SSIP configurations absorb the population formerly held by CIP.
5. $p_{\max}$ (the dominant SSIP fraction) rises.
6. n_{sig} (number of meaningfully populated configurations) falls.
7. SCE falls from $\sim$1.62 toward $\text{SCE}^* = 1.466$.

The process is dose-dependent and monotone in SCE until TMB is present in sufficient excess to fully suppress CIP. Beyond that point, further TMB addition has diminishing effect on SCE (no CIP remains to suppress) but may introduce TMB–DME or other secondary interactions that could affect conductivity. The correct TMB dose is the minimum needed to reach SCE ≈ 1.466, not the maximum tolerable.

4.3 Predicted Performance at SCE* = 1.466

From the confirmed framework equations (Preprint 1, pre-registered at https://github.com/Eric-Robert-Lawson/attractor-oncology):

Regime 1 RT gradient (Equation 1):

$$\ln(100 - \mathrm{CE}_{\mathrm{RT}} + 0.1) = 1.493 + 1.230 \cdot \mathrm{SCE} \tag{3}$$

LT band (Equation 2):

$$\mathrm{CE}_{\mathrm{LT}} = 11.91 + 33.21 \cdot \mathrm{SCE} \tag{4}$$

At SCE = 1.466:

$$\mathrm{CE}_{\mathrm{RT}}^{\mathrm{pred}} = 100.1 - e^{1.493+1.230\times 1.466} = 100.1 - 27.0 = 73.1\% \tag{5}$$

$$\mathrm{CE}_{\mathrm{LT}}^{\mathrm{pred}} = 11.91 + 33.21 \times 1.466 = 60.6\% \tag{6}$$

Remark 4.1. These predictions represent the geometry-driven component of performance at the fixed point — the floor set by solvation structure alone, before salt optimisation, additive tuning, or electrode pre-treatment. The key experimental test is not the absolute values but the *direction*: $\mathrm{CE}_{\mathrm{RT}}$ must rise and $\mathrm{CE}_{\mathrm{LT}}$ must not fall as TMB is added and SCE is reduced from ∼1.62 toward 1.466. Any formulation in which both improve simultaneously is consistent with the framework. Any formulation in which RT improves but LT collapses has overshot the fixed point into Failure Mode B.

4.4 The Target Solvation Structure

At $\mathrm{SCE}^* = 1.466$ with $k = 3$ significant configurations, the target population is (from Derivation 2 of the pre-registered record):

Table 3: Target Li^+ first-shell coordination geometry population at $\mathrm{SCE}^* = 1.466$ for the DOL/TMB system. CIP population is suppressed to the minority tertiary configuration. Compare to the DOL baseline where CIP is a secondary contributor.

Configuration	Target fraction	Coordination type	Change from baseline
Dominant (p_1)	≈ 0.38	SSIP (DOL-only)	Rises from baseline
Secondary (p_2)	≈ 0.38	SSIP variant	Rises from baseline
Tertiary (p_3)	≈ 0.24	CIP (suppressed)	Falls from baseline
SCE	1.466	$-\sum p_i \ln p_i$	Falls from ∼1.62

The defining feature of the target state is the suppression of CIP to the tertiary position: $p_{\mathrm{CIP}} \approx 0.24$, down from ∼0.20–0.30 at the DOL baseline (where it is a secondary contributor, not tertiary). This is precisely what TMB achieves by intercepting FSI^- preferentially.

5 Candidate Formulation and Falsification Conditions

The following candidate is derived from the anion receptor mechanism and pre-registered in the public repository at https://github.com/Eric-Robert-Lawson/attractor-oncology. The timestamped commit history is the priority record.

5.1 Candidate 3: LiFSI in DOL/TMB

Formulation: LiFSI 1.0 M in DOL/TMB 4:1 v/v (starting ratio; to be confirmed by SCE measurement).

Target SCE: 1.466 ± 0.05.

Mechanistic basis: TMB coordinates FSI^- preferentially, reducing the free anion activity in solution, suppressing the CIP population in the Li^+ primary shell, and contracting the coordination geometry distribution from above SCE* toward the fixed point. TMB does not coordinate Li^+ and does not introduce a new coordination class.

Experimental protocol:

1. Prepare LiFSI 1.0 M solutions in DOL/TMB at volume ratios: DOL:TMB = 9:1, 6:1, 4:1, 3:1.
2. Measure Raman spectrum of each solution. Track:
   - FSI^- S–N stretching mode (≈ 740 cm^{-1}): shifts on CIP formation vs. SSIP; used to estimate p_{CIP}.
   - B–O stretching mode of TMB (≈ 955 cm^{-1}): shifts on TMB–FSI^- complex formation; confirms anion receptor activity.
   - DOL ring breathing mode (≈ 940 cm^{-1}): monitors DOL coordination to Li^+.
3. Estimate p_{CIP} from FSI^- Raman peak shift and compute SCE. Identify the ratio at which SCE ≈ 1.466.
4. Measure $\mathrm{CE_{RT}}$ at 25 °C and $\mathrm{CE_{LT}}$ at -20 °C at the target ratio: 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2}, 1.0 mAh cm^{-2}.
5. Apply falsification conditions below.

Performance predictions (at SCE = 1.466):

$$\mathrm{CE}_{\mathrm{RT}}^{\mathrm{pred}} = 73\% \text{ (geometry-driven floor, Regime 1)}$$
$$\mathrm{CE}_{\mathrm{LT}}^{\mathrm{pred}} = 61\% \text{ (LT band)}$$

Falsification conditions:

- **Primary falsification (directional):** Framework falsified if $\mathrm{CE_{RT}}$ does *not* improve monotonically as TMB fraction increases from 0 toward the target ratio while $\mathrm{CE_{LT}}$ is maintained. Both must not decline simultaneously in any direction inconsistent with the population contraction prediction.
- **Secondary falsification (absolute, at confirmed SCE):** Framework falsified if at the TMB ratio confirmed by Raman to achieve SCE ≈ 1.466: $\mathrm{CE_{RT}} < 87\%$ OR $\mathrm{CE_{LT}} < 60\%$.
- **Mechanistic falsification (anion receptor activity):** Anion receptor mechanism falsified if Raman shows *no* B–O shift on TMB addition (TMB–FSI^- complex not forming) OR no change in FSI^- S–N mode (CIP population not responding to TMB addition).
- **Overcorrection detection (Failure Mode B):** If $\mathrm{CE_{RT}}$ improves substantially (e.g., $> 5\%$ absolute) while $\mathrm{CE_{LT}}$ falls by more than 5% absolute, Failure Mode B is diagnosed: TMB has driven SCE below SCE*. The correct response is to reduce the TMB fraction and re-measure, not to conclude that the framework is falsified.

5.2 Dose Sensitivity and the TMB Optimum

The relationship between TMB molar fraction x_{TMB} and the resulting SCE is expected to be monotonically decreasing in the regime where CIP suppression is incomplete:

$$\frac{d(\mathrm{SCE})}{dx_{\mathrm{TMB}}} < 0 \quad \text{for } x_{\mathrm{TMB}} < x_{\mathrm{TMB}}^{\mathrm{sat}} \tag{7}$$

where $x_{\mathrm{TMB}}^{\mathrm{sat}}$ is the saturation point at which all CIP has been suppressed and further TMB addition has no additional effect on SCE. The target ratio of 4:1 (DOL:TMB) is estimated to fall below saturation, leaving room for fine adjustment. The ratio scan (9:1, 6:1, 4:1, 3:1) is designed to bracket the optimum and identify x_{TMB} at which SCE ≈ 1.466.

6 Molecular Search: TMB as the Canonical Boron Ester Anion Receptor

6.1 Search Protocol

A systematic molecular search was conducted in the MU (Molecular Universe) database using SMILES-based neighbourhood exploration anchored on the boron ester class. The full search record is archived at https://github.com/Eric-Robert-Lawson/attractor-oncology, file `MU_Molecular_Search_Comple`

The search criteria for a valid anion receptor were:

(i) Lewis acidic boron centre (or alternative Lewis acid centre capable of FSI^- coordination).

(ii) No lone-pair donor sites capable of coordinating Li^+ (boron esters satisfy this; boron ethers do not).

(iii) Liquid at $-20\,°\mathrm{C}$.

(iv) Commercially available.

(v) Electrochemically stable at Li metal potentials.

6.2 TMB as the Canonical Anion Receptor

TMB is the smallest, most symmetric, and most accessible boron ester. It satisfies all five search criteria simultaneously. Its boron centre is Lewis acidic via the empty p orbital, and its three methoxy groups are electron-donating by induction but not sufficiently so to block Lewis acid activity toward FSI^-.

Larger boron esters (triethyl borate TEB, tributyl borate TBuB) satisfy criteria (i) and (ii) but have higher viscosity and lower commercial availability at battery-grade purity. Their Lewis acidity is comparable to TMB; the dose required to achieve the same CIP suppression is similar on a molar basis. They are secondary candidates if TMB proves incompatible for application-specific reasons (boiling point $68.7\,°\mathrm{C}$ may be a concern in some cell formats).

Non-boron Lewis acids explored in the search (aluminium esters, silicon esters) either coordinate Li^+ (violating criterion ii) or are insufficiently Lewis acidic for FSI^- (fluoride ion affinity below the relevant threshold). No non-boron candidate satisfies all five criteria simultaneously.

Proposition 6.1 (TMB as canonical boron ester anion receptor)**.** *Within the accessible commercial solvent/additive space explored by the molecular search, TMB is the unique molecule satisfying* *all five anion receptor design criteria simultaneously for application in LiFSI-based electrolytes at* $-20\,^\circ C$*. TEB and TBuB are identified as secondary candidates with comparable Lewis acidity but inferior physical properties for the target application.*

6.3 The Boron Ester Class as Region 3 of the Molecular Universe

The molecular search record establishes three regions of the battery-relevant molecular universe accessible from ether-anchored searches:

Region 1 — Ether class: Linear and cyclic ethers. Primary solvents. SCE range: ∼1.27–1.68.

Region 2 — Phosphate ester class: TMP and analogues. Cross-class ESP-matched co-solvents. SCE raising mechanism.

Region 3 — Boron ester class: TMB and analogues. Anion receptors. SCE lowering mechanism.

These three regions are geometrically distinct in chemical space and serve distinct functional roles in the SCE Framework. Region 3 (boron esters) is the only class that operates exclusively on the anion rather than on the solvent geometry. Its role in the framework is therefore complementary to and non-overlapping with Regions 1 and 2.

7 Geometric Asymmetry: Anion Reception and ESP Matching as Orthogonal Levers

7.1 The Two Levers Act on Different Components

ESP matching (the mechanism for approaching SCE* from below) and anion reception (the mechanism for approaching SCE* from above) are orthogonal in a precise geometric sense:

Table 4: Geometric comparison of the two mechanisms for navigating to SCE* = 1.466. The mechanisms are orthogonal: they act on different components of the electrolyte and produce opposite SCE trajectories.

Property	ESP matching (from below)	Anion reception (from above)
Starting position	SCE < SCE* (too ordered)	SCE > SCE* (too disordered)
SCE trajectory	↑ toward SCE*	↓ toward SCE*
Component targeted	Co-solvent (new coordination class added)	Anion (existing coordination class removed)
Shell geometry effect	New coordination modes introduced	No new coordination modes; existing modes contracted
Population effect	$p_{\max}$ falls; n_{sig} rises	$p_{\max}$ rises; n_{sig} falls
Canonical realisation	DME/TMP	DOL/TMB
Design criterion	$\lvert\Delta\text{ESP}\rvert \leq k_BT$, distinct class	Lewis acid: FSI^- active, Li^+ inactive
Raman signature	P=O shift (TMP entering shell)	B–O shift (TMB–FSI^- complex) + FSI^- mode shift

7.2 Orthogonality and Combination

Because the two levers act on different components of the electrolyte (co-solvent geometry vs. anion activity), they can in principle be combined in a single formulation. A system containing a base ether solvent, an ESP-matched phosphate ester co-solvent, and a boron ester anion receptor simultaneously applies both levers and has access to three distinct handles on SCE: solvent class (Region 1), cross-class ESP contrast (Region 2), and anion suppression (Region 3).

This three-class architecture is the structural basis for the extreme low-temperature candidate formulations in the SCE Framework series. The combination is documented in the pre-registration record at https://github.com/Eric-Robert-Lawson/attractor-oncology and its derivation is archived in the pre-registered derivation record. The present paper establishes the second lever independently and completely, without relying on the combination.

7.3 Why the Levers Are Not Redundant

A system below SCE* cannot be correctly navigated by anion reception alone: suppressing the CIP population in a system that is already SSIP-dominant (low CIP fraction) would reduce SCE further below SCE*, worsening LT performance. The wrong lever applied in the wrong direction produces the wrong outcome.

A system above SCE* cannot be correctly navigated by ESP matching alone: adding a new coordination class to a system that is already too disordered would raise SCE further above SCE*, worsening RT performance. Again, the wrong lever in the wrong direction worsens the outcome.

The framework therefore specifies not just the target but the direction of approach and the correct lever for each direction. This directional specificity is itself a testable prediction: applying the wrong lever in the wrong direction must produce a predictable failure mode.

8 Generalisation of the Anion Receptor Principle

8.1 The General Design Rule for Systems Above SCE*

The anion receptor mechanism generalises beyond the DOL/TMB pair. The design rule for any system above SCE* is:

1. Identify the source of excess SCE: which coordination class (CIP, AGG, or mixed solvent/anion configurations) is contributing the entropy excess beyond what is present in the SSIP-dominated baseline?

2. Select an anion receptor with Lewis acidity sufficient to compete for the anion in that coordination class, but insufficient to coordinate Li^+ directly.

3. Confirm by Raman or MD that the targeted coordination class is suppressed on receptor addition.

4. Confirm by SCE measurement that the suppression moves SCE toward SCE* rather than past it.

5. Apply the falsification conditions.

8.2 Application Beyond DOL

- **2-MeTHF systems:** 2-MeTHF has SCE = 1.55 in the dataset, slightly above SCE*. If this arises from a modest CIP contribution, a small TMB addition may be sufficient to reach SCE*. If it arises from the ring conformational flexibility generating multiple SSIP variants, the anion receptor mechanism is the wrong lever and a concentration adjustment is more appropriate. The distinction is diagnosable by Raman before any CE measurement.

- **High-concentration systems:** AGG-dominant systems (Regime 2) have a qualitatively different SCE–CE relationship. Anion reception in a Regime 2 system would reduce the AGG population, potentially crossing the regime boundary and producing a regime transition rather than a smooth SCE reduction. This is a specific failure mode of the anion receptor mechanism in Regime 2 systems and constitutes a named prediction of the framework.

8.3 Extension to Other Anions and Metal Ions

The anion receptor mechanism is not FSI*-specific. For other anions ($TFSI^-$, PF_6^-, $DFOB^-$), the relevant Lewis acid must be identified by matching the fluoride or nitrogen ion affinity of the receptor to the binding energy of the anion with the target metal ion. For non-lithium chemistries (Na^+, K^+, Zn^{2+}), the same structural logic applies: identify the coordination class contributing excess entropy, select a receptor that suppresses it without entering the cation shell. The SCE* value for each metal ion chemistry is a separate derivation (each metal ion has its own confirmed gradient equations and therefore its own fixed point), but the anion receptor mechanism for approaching it from above is fully general.

9 Discussion

9.1 The Significance of the Directional Specificity

The most important result of this paper is not the candidate formulation but the directional specificity of the framework. Most electrolyte design approaches treat the problem as undirected: try different additives, measure CE, iterate. The SCE Framework assigns a direction to every electrolyte: measure SCE, determine whether you are above or below SCE*, and apply the correct lever.

This directional knowledge changes the experimental program qualitatively. It predicts that specific additives will work and specific additives will fail for specific base solvents. It predicts that the same additive (TMB) applied to a base solvent below SCE* would worsen performance rather than improve it. These are falsifiable predictions that the prior literature does not make.

9.2 The DOL/DME Negative Control Revisited

The DOL/DME 1:1 result ($CE_{RT} = 98.9\%$, $CE_{LT} = 27.5\%$) was identified in Preprint 1 as the negative control confirming Failure Mode B. The present paper adds a mechanistic interpretation: DME addition to DOL does not gently correct the excess SCE; it aggressively displaces the DOL/CIP population with a DME-dominant SSIP population, driving SCE from $\sim$1.62 to 1.27 — a correction of -0.35 when only -0.15 was needed. The system overshoots by more than twice the required correction.

This interpretation is testable: the Raman spectrum of DOL/DME 1:1 should show near-complete suppression of the DOL coordination mode and near-complete dominance of the DME bidentate SSIP configuration. That prediction is in the pre-registration record.

9.3 Why Not Just Increase Salt Concentration?

Salt concentration increase is the conventional route to improving CE_{RT} in ether-based electrolytes. It works by driving the system into AGG-dominant Regime 2, where the RT–SCE gradient inverts and high SCE produces high CE_{RT}.

The framework predicts that this route is available but costly: it requires concentrations of > 3 M, which substantially increases viscosity, reduces ionic conductivity, and raises cost. The anion receptor route achieves the same population contraction at 1 M salt concentration by targeting the anion chemistry rather than the concentration regime. The two routes are not contradictory; they address the same problem through different mechanistic pathways. The anion receptor route is the lower-concentration, lower-viscosity alternative.

9.4 Limitations

- The DOL baseline SCE ($\sim$1.60–1.65) is estimated from the dataset position and structural reasoning. Direct MD or Raman measurement of pure LiFSI/DOL coordination populations is needed to establish the exact baseline before the TMB dose can be precisely predicted.
- TMB has a low boiling point (68.7 °C), which may create vapour pressure concerns in sealed cell formats. Higher boron esters (TEB) offer a practical alternative.
- The Raman-based SCE estimation for the DOL/TMB system is more complex than for DME/TMP, because three Raman features must be tracked simultaneously (DOL, FSI^-, TMB–FSI^- complex). MD simulation is strongly preferred for quantitative SCE determination in this system.
- The falsification threshold $CE_{RT} < 87\%$ at confirmed SCE is set conservatively. The predicted geometry-driven floor is 73%; the threshold at 87% assumes modest additive and concentration optimisation. If the unoptimised DOL/TMB system does not exceed 73%, the threshold still stands and the framework is not falsified.

10 Conclusion

We have introduced the **anion receptor mechanism** as the correct intervention for approaching the Li metal battery performance fixed point $SCE^* = 1.466$ from above.

The mechanism operates as follows: when the Li^+ first-shell population is too disordered ($SCE > SCE^*$), the excess entropy arises from anion participation in the primary shell (CIP and mixed configurations). A Lewis acid anion receptor (TMB) intercepts the anion before it enters the Li^+ shell. The CIP population is suppressed. The distribution contracts. SCE falls toward SCE^*. The mechanism does not introduce a new coordination class; it removes an existing contributor.

DOL is identified as the canonical base solvent above SCE^*, and TMB is identified as the canonical anion receptor within accessible commercial chemistry. Both identifications are supported by the pre-registered molecular search record at https://github.com/Eric-Robert-Lawson/attractor-oncology.

The anion receptor mechanism is geometrically distinct from and orthogonal to the ESP-matching mechanism: the two levers act on different electrolyte components, produce opposite SCE trajectories, and are applicable to opposite sides of SCE*. Their orthogonality means they can be combined — a three-class architecture applying both levers simultaneously is derivable from the framework and documented in the pre-registration record.

Candidate 3 (LiFSI 1.0 M in DOL/TMB 4:1 v/v) is pre-registered with falsification conditions at four levels of specificity: primary (directional), secondary (absolute at confirmed SCE), mechanistic (Raman confirmation of anion receptor activity), and overcorrection detection (Failure Mode B diagnosis).

No prior work identifies excess CIP population as the source of RT performance suppression in DOL-based electrolytes and applies an anion receptor to correct it by targeted SCE reduction. The mechanism is new. The intervention is computable. The experiment is bounded and falsifiable.

Data and derivation record availability. All pre-registered predictions, falsification conditions, molecular search records, and derivations referenced in this paper are archived with full timestamped commit history at:
https://github.com/Eric-Robert-Lawson/attractor-oncology
The repository commit history constitutes the priority and pre-registration record for all claims in this paper.

Relationship to the SCE Framework preprint series. This paper is one of a series deriving from the SCE Framework. Each paper is self-contained. The complete framework, all candidate formulations, all falsification conditions, and the web of derivations connecting the papers are documented in the pre-registration record at the repository above.

Competing interests. The author declares no competing financial interests.

Author contributions. E.R.L.: conceptualisation, derivation, mechanistic analysis, writing.

A Derivation of the Anion-Mediated Entropy Contribution

Let the full coordination population before TMB addition consist of k configurations with fractions $\{p_i\}_{i=1}^{k}$, summing to unity. Let configuration k be the CIP class with fraction $p_k = p_{\mathrm{CIP}}$.

SCE before TMB:

$$\mathrm{SCE}_0 = -\sum_{i=1}^{k} p_i \ln p_i \tag{8}$$

After complete CIP suppression ($p_{\mathrm{CIP}} \to 0$), the remaining $k-1$ configurations renormalise: $p_i' = p_i/(1-p_{\mathrm{CIP}})$ for $i = 1, \ldots, k-1$. SCE after:

$$\mathrm{SCE}_1 = -\sum_{i=1}^{k-1} p_i' \ln p_i' = -\sum_{i=1}^{k-1} \frac{p_i}{1-p_{\mathrm{CIP}}} \ln \frac{p_i}{1-p_{\mathrm{CIP}}} \tag{9}$$

The SCE reduction on complete CIP suppression:

$$\begin{aligned}
\Delta\mathrm{SCE} &= \mathrm{SCE}_1 - \mathrm{SCE}_0 \\
&= -\sum_{i=1}^{k-1} \frac{p_i}{1-p_{\mathrm{CIP}}} \ln \frac{p_i}{1-p_{\mathrm{CIP}}} + \sum_{i=1}^{k} p_i \ln p_i \\
&= \ln(1-p_{\mathrm{CIP}}) + p_{\mathrm{CIP}} \ln p_{\mathrm{CIP}} \cdot \frac{1}{1-p_{\mathrm{CIP}}} \cdot (1-p_{\mathrm{CIP}}) + p_{\mathrm{CIP}} \ln p_{\mathrm{CIP}}
\end{aligned} \tag{10}$$

For $p_{\text{CIP}} = 0.20$ (estimated DOL baseline):

$$\begin{aligned}\Delta\text{SCE} &\approx \ln(0.80) + 0.20\ln(0.20)\\ &= -0.223 + 0.20 \times (-1.609)\\ &= -0.223 - 0.322 = -0.545\end{aligned}$$

This estimate (ΔSCE ≈ -0.55 on complete CIP suppression) exceeds the required correction of -0.15. Partial suppression (not complete elimination) of the CIP population is therefore sufficient to reach SCE*, consistent with the expectation that the 4:1 DOL:TMB ratio does not fully eliminate CIP but suppresses it to the tertiary position.

B TMB Lewis Acidity: Fluoride Ion Affinity Comparison

Table 5: Gas-phase fluoride ion affinity (FIA) as a proxy for Lewis acidity, compared across boron esters and reference Lewis acids. FIA values from [4] and references therein. TMB FIA is sufficient for FSI^- coordination (electronegative fluorine centres) but below the threshold for Li^+ coordination (hard cation, requires donor not acceptor).

Molecule	FIA (kJ mol^{-1})	Li^+ coordination?
BF_3	365	No (strong Lewis acid, no lone pairs)
TMB (this work)	∼340	No (confirmed)
TEB	∼335	No
$Al(OEt)_3$	∼420	Yes (too Lewis acidic; coordinates Li^+)
THF	—	Yes (Lewis base; coordinates Li^+)

The FIA of TMB (∼340 kJ mol^{-1}) places it in the range of moderate Lewis acids capable of anion coordination without the extreme acidity that would displace Li^+ from its coordination sphere or react destructively with the electrolyte.

C Pre-Registered Falsification Conditions (Verbatim)

The following conditions are reproduced from the repository commit record. The commit timestamp predates this document.

Candidate 3 (LiFSI 1.0M, DOL/TMB ratio scan):

- *Primary:* Framework falsified if CE_{RT} does not improve and CE_{LT} is not maintained as TMB fraction increases from 0 toward the target ratio.
- *Secondary:* Framework falsified if at the ratio confirmed by Raman to have SCE ≈ 1.466: $CE_{RT} < 87\%$ OR $CE_{LT} < 60\%$.
- *Mechanistic:* Anion receptor mechanism falsified if Raman shows no B–O shift on TMB addition OR no change in the FSI^- S–N stretching mode.
- *Overcorrection:* Failure Mode B diagnosed (not framework falsification) if CE_{RT} improves $> 5\%$ absolute while CE_{LT} falls $> 5\%$ absolute. Correct response: reduce TMB fraction.

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